

RESEARCH ARTICLE | JANUARY 12 2026

Accurate learning of long-range interatomic potentials by coupling Cartesian atomic cluster expansion and sum-of-Gaussians neural networks

Yajie Ji ; Jiuyang Liang ; Zhenli Xu 



J. Chem. Phys. 164, 024109 (2026)

<https://doi.org/10.1063/5.0303312>



Articles You May Be Interested In

Long-range electrostatics for machine learning interatomic potentials is easier than we thought

J. Chem. Phys. (February 2026)

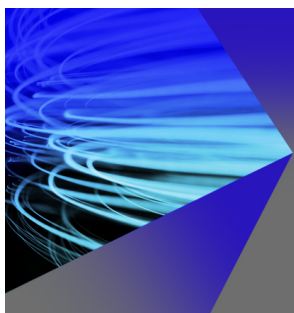
Improved loss functions for machine-learned atomic potentials

J. Chem. Phys. (October 2025)

Incorporating Coulomb interactions with fixed charges in moment tensor potentials and equivariant tensor network potentials

J. Chem. Phys. (February 2026)

12 March 2026 01:18:47



AIP Advances

Why Publish With Us?



21DAYS
average time
to 1st decision



OVER 4 MILLION
views in the last year



INCLUSIVE
scope

[Learn More](#)

 AIP
Publishing

Accurate learning of long-range interatomic potentials by coupling Cartesian atomic cluster expansion and sum-of-Gaussians neural networks

Cite as: J. Chem. Phys. 164, 024109 (2026); doi: 10.1063/5.0303312

Submitted: 20 September 2025 • Accepted: 22 December 2025 •

Published Online: 12 January 2026



Yajie Ji,^{1,2,a)} Jiuyang Liang,^{1,3,b)} and Zhenli Xu^{1,c)}

AFFILIATIONS

¹ School of Mathematical Sciences, MOE-LSC and CMA-Shanghai, Shanghai Jiao Tong University, Shanghai 200240, China

² Department of Statistics and Data Science, Yale University, New Haven, Connecticut 06511, USA

³ Center for Computational Mathematics, Flatiron Institute, Simons Foundation, New York, New York 10010, USA

^{a)}jiyajie595@sjtu.edu.cn

^{b)}Author to whom correspondence should be addressed: jliang@flatironinstitute.org

^{c)}xuzl@sjtu.edu.cn

ABSTRACT

Accurately capturing long-range interactions is critical for molecular dynamics simulations based on machine learning interatomic potentials. We recently proposed the sum-of-Gaussians neural network (SOG-Net), which learns long-range energy contributions directly from energy and force data such that the long-range tail of different decay rates can be well fitted. In this work, we incorporate the SOG-Net with a short-range descriptor of the Cartesian atomic cluster expansion, resulting in the CACE-SOG model, to show that the SOG-Net is a general module that can be coupled with different short-range descriptors. We also study new technical developments in the SOG-Net, including improved extrapolation accuracy, handling of different charge states, and faster convergence. We evaluated the CACE-SOG model across a diverse set of systems, including molecular dimers, aqueous salt solutions, charged ionic clusters, and liquid-vapor and Pt(111) interfacial water systems, and compared it with the CACE-based latent Ewald summation and the CACE-only methods. These results demonstrate that the SOG-Net is promising for accurately learning long-range interatomic interactions.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0303312>

I. INTRODUCTION

Machine learning interatomic potentials (MLIPs) have introduced a new paradigm for all-atom molecular dynamics simulations, offering the opportunity to avoid expensive quantum mechanical calculations while preserving accuracy levels as in *ab initio* molecular dynamics.^{1,2} Accurate modeling of long-range (LR) interactions remains a fundamental challenge in MLIPs, as most existing models are built on local atomic environments and do not adequately account for LR interactions that decay slowly with distance,^{3,4} including Coulomb, dispersion, and dipole-dipole interactions. LR interactions play an essential role in determining the structure, dynamics, and thermodynamic properties of complex systems such as electrochemical interfaces, electronically driven phase transitions, protein folding, and polar materials.^{5–9}

Various approaches have been developed in the literature in order to incorporate LR effects into MLIPs. Improving local models

with empirical corrections for electrostatics and dispersion^{10,11} allows us to take into account these LR effects, but this idea remains a difficulty for its transferability to different environments. The fourth-generation high-dimensional neural network potentials (4G-HDNNPs)⁶ propose to predict effective atomic partial charges while enforcing global charge conservation by using the charge equilibration scheme.¹² In the Deep Potential (DP) LR model¹³ and self-consistent field neural networks,¹⁴ it was suggested to learn virtual charges at positions of maximally localized Wannier centers. Once these charges are learned, electrostatic interactions are computed using the fast Fourier transform (FFT)-based Ewald summation,^{15,16} which enables fast LR force evaluation. The long-distance equivariant method^{7,17} is an alternative approach that does not rely on explicit charge assignment but incorporates LR effects via local representations and an LR message-passing model.¹⁸ This is achieved by coupling with learnable frequency filters in reciprocal space. By the

“nearsightedness principle”¹⁹ rooted in electronic structure theory, the latent Ewald summation (LES) method²⁰ also provides an elegant approach that learns the so-called “latent variable” $\mathbf{q}$ from local atomic features. Different from previous works that explicitly learn charges, $\mathbf{q}$ in the LES can be high dimensional to represent parameters for more generalized LR interactions. These $\mathbf{q}$ then interact also via the Ewald summation to determine the global energy.

Despite these advances, there are many open problems for learning LR interactions.²¹ Several studies have shown that LR interactions in realistic systems often deviate from idealized power-law decay.^{4,7} This discrepancy arises in part because atomic charges in density functional theory (DFT) reflect spatially distributed electron densities, rather than point-like sources. The widely used Ewald summation, originally designed for $1/r$ Coulombic interactions, is not well suited to capture generalized LR decay rates. Recently, we proposed the sum-of-Gaussians neural network (SOG-Net)²² to learn LR interactions with different decay tails. Similar to the LES, the SOG-Net predicts a latent variable $\mathbf{q}$ from local atomic features using an embedded neural network,²³ without relying on predefined atomic charges. Differently, the SOG-Net introduces a reciprocal-space convolution layer where the Fourier multiplier is expressed as a sum of Gaussians with trainable amplitudes and bandwidths. This design allows capturing a broad range of LR interaction profiles while retaining near-linear computational scaling through the use of non-uniform fast Fourier transforms (NUFFT).^{24,25} Although the SOG-Net is, in principle, compatible with various short-range (SR) MLIP architectures,^{26–29} it has so far been implemented with the DP descriptor.²³ Performance evaluations using alternative SR descriptors, such as graph neural networks (GNNs) and body-order expansion-based descriptors, remain unexplored.

In this paper, we extend the SOG-Net framework by integrating it with the Cartesian atomic cluster expansion (CACE) descriptor,³⁰ resulting in a new CACE-SOG model. The CACE is a message-passing, graph-based, explicit SR descriptor that is invariant to rotation, translation, and permutation. Compared with the fully data-driven DP descriptor, explicit descriptors such as ACE and CACE often require fewer trainable parameters and provide higher data efficiency.^{31,32} Relative to ACE, CACE offers a mathematically well-defined and more efficient route for MLIP design by working in Cartesian coordinates while preserving rotational symmetry and body orders; it is also straightforward to implement and to extend with new features. Because SOG-Net uses the SR descriptor to learn latent variables, pairing it with explicit CACE descriptors can improve both training and simulation efficiency while maintaining accuracy.

To our knowledge, current LR models built on such explicit SR descriptors rely on deterministic Ewald summation;³³ we are not aware of a model that couples them with a data-driven LR module such as SOG-Net. Recently, the CACE-based latent Ewald summation (CACE-LES)²⁰ has been shown to capture electrostatics and to recover effective charges and polarization effects from energy/force data.^{34,35} Replacing Ewald summation with SOG-Net allows more general LR effects within a unified framework while keeping data efficiency. We validate the performance of the CACE-SOG, demonstrating that the model inherits the data efficiency and physical consistency of graph-based neural networks while introducing a flexible and learnable mechanism to capture non-Coulombic LR interactions. Importantly, for systems with Coulomb interactions,

the CACE-SOG achieves similar accuracy as the CACE-LES. When the system includes non-Coulombic LR interactions, the CACE-SOG shows strong superiority. The accuracy of the new model is similar to the DP-based SOG-Net²² but with fewer trainable parameters, further demonstrating that the SOG-Net is a general framework for LR learning. These findings suggest that the SOG-Net is promising for modeling LR interactions by incorporating existing SR descriptors in a uniform framework.

II. METHODS

Consider an atomic system of N atoms located at $\mathbf{r}_1, \dots, \mathbf{r}_N$ in the domain Ω with periodic boundary conditions. The total potential energy is decomposed into SR and LR components: $E = E^{\text{sr}} + E^{\text{lr}}$. The SR energy E^{sr} is expressed as a sum over atomic contributions,

$$E^{\text{sr}} = \sum_{i=1}^N E_{\text{SR},i} = \sum_{i=1}^N f_{\theta_{\text{SR}}}(\mathbf{D}_i), \quad (1)$$

where $\mathbf{D}_i$ is the local atomic environment descriptor for atom i , and $f_{\theta_{\text{SR}}}(\cdot)$ is a fully connected neural network parameterized by θ_{SR} that maps the descriptor to the atomic energy. The LR energy E^{lr} is often modeled in Fourier space. In this work, we introduce the CACE and the SOG-Net to treat SR and LR components, respectively, by making an explicit connection between the two networks.

A. Short-range descriptor

The SR descriptors have been broadly studied in Refs. 23, 26–29, and 36–38, and here we adopt a newly developed CACE by Cheng³⁰ as the SR descriptor for constructing the machine-learning potential of LR systems. Different from the ACE²⁹ and most of the other E(3) equivariant MPNNs that rely on spherical harmonics as an angular basis, the CACE encodes angular dependence in a Cartesian basis, avoiding the use of spherical transformations, tensor products, and Clebsch–Gordan matrices by the Cartesian atomic cluster expansion. It provides a simple and efficient way to describe many-body interactions such that it is invariant to rotation, translation, and permutation. These advantages motivate our choice of the CACE as the SR descriptor, in spite of the fact that the network structure presented in this work can, in principle, be combined with any of the existing SR descriptors.

In the CACE descriptor, each local atomic environment is represented as a graph, with atoms as nodes. For neighbors within a cutoff radius r_{cut} , directed edges connect each neighbor j to the central atom i . Each atomic state is defined as $\sigma_i = (\mathbf{r}_i, \boldsymbol{\theta}_i, \mathbf{h}_i)$, with $\mathbf{r}_i$ being the Cartesian position, $\boldsymbol{\theta}_i$ being an embedding of the chemical element, and $\mathbf{h}_i$ being the learnable features. To construct $\boldsymbol{\theta}_i$, the element α_i is first encoded as a one-hot vector of length N_{element} , which is then mapped by a learnable weight matrix of size $N_{\text{embedding}} \times N_{\text{element}}$ to obtain the continuous embedding of length $N_{\text{embedding}}$. The atom-centered basis is obtained by summing over all incoming edges,

$$A_{i,cn\mu} = \sum_{j \in \mathcal{N}(i)} \chi_{cn\mu}(\sigma_i, \sigma_j), \quad (2)$$

where $\mathcal{N}(i)$ is the neighbor set of i . The edge basis $\chi_{cn\mu}$ is indexed by c (edge channel), n (radial basis), and μ (angular momentum).

It encodes the spatial arrangement of atom j around atom i as the product of an edge-type, a radial, and an angular bases,

$$\chi_{c\mu}(\sigma_i, \sigma_j) = T_c(\sigma_i, \sigma_j) R_{n,c\mu}(r_{ji}) \mathcal{L}_\mu(\tilde{\mathbf{r}}_{ji}), \quad (3)$$

where $\tilde{\mathbf{r}}_{ji} = \mathbf{r}_{ji}/r_{ji}$ with $\mathbf{r}_{ji} = \mathbf{r}_j - \mathbf{r}_i$ and $r_{ji} = |\mathbf{r}_{ji}|$. The edge-type basis T_c depends on the chemical elements and, if message-passing layers are used, also on the hidden features, with c labeling the edge feature dimension. The radial basis $R_{n,c\mu}$ is a set of n trainable Bessel functions multiplied by a smooth cutoff function. Let $\tilde{\mathbf{r}}_{ji} = (x_{ji}, y_{ji}, z_{ji})$, and $\mu = (\mu_x, \mu_y, \mu_z)$. The angular basis is defined as

$$\mathcal{L}_\mu(\tilde{\mathbf{r}}_{ji}) = x_{ji}^{\mu_x} y_{ji}^{\mu_y} z_{ji}^{\mu_z}, \quad (4)$$

with $\mu_x + \mu_y + \mu_z = \mu$ being the total number of angular momenta. Once the atom-centered basis is constructed, it is symmetrized to produce invariant features $\mathbf{D}_i$ with a given body order v such that v -body correlations are involved in the features. The series is truncated at $\mu_{\max}$ and $v_{\max}$, depending on the requirement of accuracy and efficiency.

By construction, $\mathbf{D}_i$ forms a complete basis of rotation-, translation-, and permutation-invariant functions of the local environment, which is input into a fully connected network for training, such that the network can describe the SR interactions of atomic systems.

B. Long-range descriptor

To incorporate LR interactions, we use another neural network $f_{\theta_q}(\cdot)$ that maps the same local descriptor $\mathbf{D}_i$ to a latent variable $\mathbf{q}_i = (q_{i,1}, \dots, q_{i,P})$ of dimension P for each atom,

$$\mathbf{q}_i = f_{\theta_q}(\mathbf{D}_i). \quad (5)$$

The underlying physical principle is the nearsightedness principle rooted in electronic structure theory,¹⁹ which states that local electronic properties, such as the electron density, are strongly influenced only by the effective potential in the nearby environment. Although the principle is empirical, it has been shown to be effective in many applications.³ The latent variable $\mathbf{q}_i$ is multi-dimensional and is designed to encode features relevant to LR interactions. When $\mathbf{q}_i$ is restricted to a one-dimensional scalar, previous work has shown that it can be interpreted as an effective atomic charge when used as input to the Ewald summation for computing electrostatic interactions.³⁴

Now we consider how the latent variable $\mathbf{q}_i$ is utilized to compute LR energy contributions. Suppose that L_x , L_y , and L_z are the system sizes along the three directions. In our SOG-Net, by leveraging the rapid decay of LR interactions in Fourier space, we account for the LR energy by using multiple convolution layers,

$$E^{\text{lr}} = \frac{1}{2V} \sum_{\eta=1}^P \sum_{\mathbf{k}} \hat{f}_{\theta_{\text{SOG}}}(\mathbf{k}) |\mathbf{S}(\mathbf{k}, \eta)|^2, \quad (6)$$

where $\mathbf{k} = 2\pi(m_x/L_x, m_y/L_y, m_z/L_z)$ denotes the Fourier modes and runs over all integers m_x , m_y , and m_z , and the structure factor $\mathbf{S}(\mathbf{k}, \eta)$ is defined as

$$\mathbf{S}(\mathbf{k}, \eta) = \sum_{i=1}^N q_{i,\eta} e^{-i\mathbf{k} \cdot \mathbf{r}_i}, \quad (7)$$

with i being the imaginary unit. $\hat{f}_{\theta_{\text{SOG}}}(\mathbf{k})$ is the Fourier multiplier parameterized as a sum of Gaussians,

$$\hat{f}_{\theta_{\text{SOG}}}(\mathbf{k}) = \sum_{\ell=0}^{M-1} w_\ell e^{-k^2/s_\ell^2}, \quad (8)$$

where $\theta_{\text{SOG}} = \{w_\ell, s_\ell\}_{\ell=0}^{M-1}$ is the set of trainable parameters. If $P > 1$ for the latent variable, θ_{SOG} can also be multilayer, which serves as an approximation of the LR tail in Fourier space. The form of the LR energy, as constructed above, is also inherently invariant to rotation, translation, and permutation.

Figure 1 illustrates a sketch map of the network architecture of the CACE-SOG model, where the CACE descriptor is used for both the SR energy and the latent variable, and the SOG-Net is used to generate the Fourier multiplier. The architecture is similar to the network in Ji *et al.*,²² but with the CACE as the SR descriptor. To accelerate the evaluation, a fast algorithm for SOG-Net can be embedded in the training procedure and molecular simulations, which maps discrete latent variables onto a uniform grid, applies an FFT, performs diagonal scaling, and then applies an IFFT. The resulting field is subsequently remapped to the positions of the variables. This algorithm follows our previous work²² and can be interpreted as a type-1 NUFFT followed by a type-2 NUFFT.²⁵ The FINUFFT package³⁹ is used in practical implementation.

The CACE-SOG model allows for physically meaningful interpretations and the extraction of quantities such as partial charges, dipole moments, and quadrupole moments under specific conditions. In particular, when the latent variable $\mathbf{q}_i$ is restricted to a single layer, its learned values can be interpreted physically. For instance, $\mathbf{q}_i$ may represent atomic partial charges or dispersion coefficients if the LR component of the potential energy surface is dominated by classical electrostatics ($1/r$) or dispersion interactions ($1/r^6$). In this context, the corresponding SOG multiplier $\hat{f}_{\theta_{\text{SOG}}}$ serves as an approximation of the LR interaction kernel in Fourier space. In quantum-mechanical systems, such as those based on DFT, the combination of the latent variable and the Fourier multiplier in the CACE-SOG model can be viewed as a coarse-grained representation of electron density polarization. In both classical and quantum settings, if the latent variable is restricted to a single layer, it can be used to compute dipole and quadrupole moments. With multiple latent layers, the model captures more complex LR decays, but extracting charges and multipoles becomes difficult. It reflects a trade-off between interpretability and accuracy: although explicit multipole extraction becomes less feasible, the model gains flexibility to achieve higher accuracy in energy and force predictions. This design allows the SOG-Net to learn a wide range of interaction patterns without explicitly fitting predefined physical laws or multipole expansions.

Charge conservation is an important issue in simulations. In the Ewald-type method, the net charge can be treated as a uniform background with the zero-frequency term being removed to ensure the well-posedness of the energy. Such removal is not necessary in the CACE-SOG model, as the SOG multiplier is smooth in both real and Fourier spaces. From another perspective, the latent variable $\mathbf{q}_i$ in the CACE-SOG is not limited to representing partial charges and may follow more general rules. Therefore, the constraint $\sum q_{i,\eta} = 0$ is not strictly required. In all our tests, we did not observe loss of accuracy or artifacts from omitting this constraint.

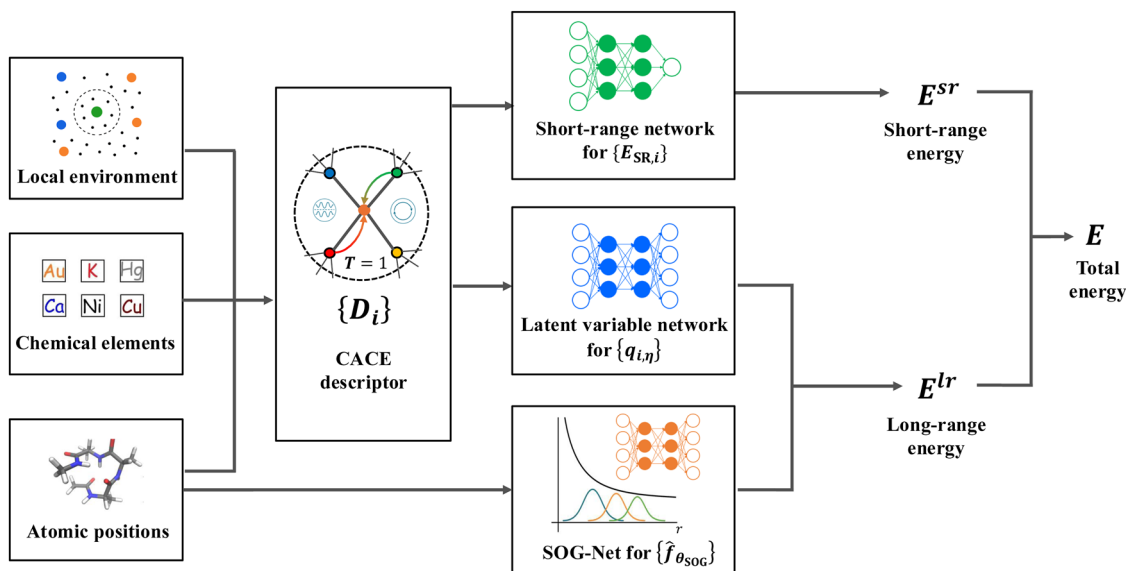


FIG. 1. Schematic of the network structure of the CACE-SOG model. Here, $\{q_i\}$, $\{D_i\}$, E^{sr} , and E^{lr} represent the latent variable, CACE descriptor (illustrated with one message-passing layer, $T = 1$), short-range energy, and long-range energy, respectively.

In contrast, for the MLIP models that explicitly learn charges, such as 3G-HDNNP, the lack of charge equilibration may result in dramatically larger errors.⁶ If needed, neutrality can be enforced for CACE-SOG by adding a penalty term to the total potential energy, $E_C = \lambda(Q - \sum q_{i,\eta})^2$, where $\lambda > 0$ is a Lagrangian multiplier and Q is the referenced quantity of the system, enforcing global conservation of the desired quantity.

Another challenge is handling different charge states. In CACE-SOG and most MLIPs, the local descriptor $\{D_i\}$ does not encode the net charge or oxidation state. It can cause problems when handling systems with varying charge states simultaneously.^{6,34} To resolve this, we adopt the strategy of King *et al.*:³⁴ when the training set contains multiple net charges, we concatenate the total net charge Q to the local descriptor D_i and use $D_i \oplus Q$ as input to predict the short-range energy and latent variable,

$$E^{sr} = \sum_{i=1}^N f_{\theta_{sr}}(D_i \oplus Q) \text{ and } q_i = f_{\theta_q}(D_i \oplus Q), \quad (9)$$

without modifying the SOG-Net architecture. The predicted q_i are then substituted into Eq. (6) to compute the long-range energy. The above strategy makes the model aware of the system charge and can represent potential energy surfaces associated with different charge states.

C. Initialization in training procedure

In our previous work,²² a simple initialization strategy was used for θ_{SOG} . In particular, the weights $\{w_\ell\}$ were set to unity, and the bandwidths $\{s_\ell\}$ were initialized as a logarithmically spaced sequence. This heuristic approach can lead to significant deviations from the true decay behavior. Here, we propose a more effective initialization scheme based on prior knowledge of the expected LR

decay. In particular, we assume a plausible power-law form close to the target interaction and use the parameters derived from an SOG approximation to the corresponding kernel as the initial guess. By incorporating physical insight into the initialization, this method improves the representation of LR interactions and substantially reduces the training time.

We adopt the bilateral series approximation (BSA),⁴⁰ which starts from an integral representation of a power-law kernel,

$$\frac{1}{r^{2\alpha}} = \frac{1}{\Gamma(\alpha)} \int_{-\infty}^{\infty} e^{-e^{\tau} r^2 + \alpha \tau} d\tau, \quad (10)$$

where $\Gamma(\cdot)$ denotes the Gamma function. In particular, the case $\alpha = 1/2$ corresponds to the Coulombic kernel $1/r$. Applying the trapezoidal rule to Eq. (10) with quadrature points $x_m = t_0 + m\zeta$, where $t_0 = -\log(2s^2)$ and $\zeta = 2\log b$ for $s > 0$ and $b > 1$, yields the BSA of the $1/r$ kernel,

$$\frac{1}{r} \approx \mathcal{F}_{SOG}(r) = \frac{2\log b}{\sqrt{2\pi s^2}} \sum_{m=-\infty}^{\infty} b^{-m} e^{-\frac{r^2}{2b^2 m^2}}, \quad (11)$$

where terms with $m < 0$ capture the SR contributions, while terms with $m \geq 0$ correspond to the LR components. One important feature of Eq. (11) is that the pointwise relative error has the asymptotic bound as $b \rightarrow 1$ for all $r > 0$,⁴⁰

$$\mathcal{E}_r := \frac{|1/r - \mathcal{F}_{SOG}(r)|}{|1/r|} \leq 2\sqrt{2}e^{-\frac{r^2}{2\log b}}. \quad (12)$$

Once the infinite sum in the right-hand side of Eq. (11) is truncated at $0 \leq m \leq M-1$, the resulting SOG, $\mathcal{F}_{SOG}^M$, is an approximation of the LR component of $1/r$ in real space. In our MLIP training, we set $b = 2$ and $s = r_{\text{cut}}/1.989253\,683908\,0267$, so that $\mathcal{E}_r \leq 0.001$ for

$r > r_{\text{cut}}$. The double-precision constant is chosen so that $\mathcal{F}_{\text{SOG}}^M(r_{\text{cut}}) = 1/r_{\text{cut}}$ holds exactly when $b = 2$; this enforces continuity at the cutoff and reduces the approximation error.^{41,42} We then compute the Fourier transform of $\mathcal{F}_{\text{SOG}}^M$ in closed form,

$$\hat{\mathcal{F}}_{\text{SOG}}^M(\mathbf{k}) = \sum_{m=0}^{M-1} 4\pi b^{2m} s^2 \log b e^{-b^{2m} s^2 k^2/2}, \quad (13)$$

which is again a sum of Gaussians. We use this to initialize the Fourier multiplier $\hat{f}_{\theta_{\text{SOG}}}$ in Eq. (8) by setting the initial weights and bandwidths to

$$w_\ell = 4\pi b^{2\ell} s^2 \log b, \quad s_\ell = \sqrt{\frac{2}{b^{2\ell} s^2}}, \quad \ell = 0, \dots, M-1, \quad (14)$$

i.e., $\theta_{\text{SOG}} = \{w_\ell, s_\ell\}_{\ell=0}^{M-1}$. Detailed derivation of Eqs. (10)–(13) is provided in Sec. I of the [supplementary material](#).

At the start of training, we choose M large enough to keep the truncation error of the Coulomb-energy tail below a target tolerance ε ,

$$\left| \sum_{\mathbf{n} \in \mathbb{Z}^3} \sum_{\substack{i,j=1 \\ |\mathbf{r}_{ji}^{\mathbf{n}}| > r_{\text{cut}}}}^N q_i q_j \left(\frac{1}{|\mathbf{r}_{ji}^{\mathbf{n}}|} - \mathcal{F}_{\text{SOG}}^M(|\mathbf{r}_{ji}^{\mathbf{n}}|) \right) \right| \leq \varepsilon \left| \sum_{\mathbf{n} \in \mathbb{Z}^3} \sum_{i,j=1}^N \frac{q_i q_j}{|\mathbf{r}_{ji}^{\mathbf{n}}|} \right|, \quad (15)$$

where $\mathbf{r}_{ji}^{\mathbf{n}} := \mathbf{r}_{ji} + \mathbf{nL}$ and the $\mathbf{n}$ -loops cover all periodic images. By default, we take $M = 12$, which in our tests keeps the LR-tail energy error at $\varepsilon = 10^{-3}$. Higher accuracy is achievable by choosing b closer to 1, but this typically requires finer Fourier grids and a larger M ,⁴² leading to higher computational cost. This physics-guided initialization acts as a simple “normalization” prior on the LR multipliers, making the initial SOG multiplier closer to the true LR decay and, thus, speeding up training and helping prevent overfitting.

If a different inverse power-law kernel besides $1/r$ is desired for the initial guess of $f_{\theta_{\text{SOG}}}$, one can discretize the integral in Eq. (10) with the corresponding α in the same manner. For simplicity, we initialize the SOG parameters using the BSA for the $1/r$ kernel in all tests reported in this work, and we find this sufficient. More generally, we suggest $1/r$ -based BSA initialization as a default setup, since electrostatics are common in LR systems and the $1/r$ BSA provides a physically reasonable LR decay compared with naïve or random initialization.

After training, we apply model reduction⁴³ to find the optimal number of Gaussians required to approximate the trained SOG multiplier within a prescribed tolerance. In practice, M can often be cut by about half without loss of accuracy. We have included a brief description of the model reduction procedure in Sec. II of the [supplementary material](#). For details on the reduction of the SOG series, see Refs. 43–46. A Python implementation of the model reduction procedure is included in our SOG-Net package (https://github.com/DukhtigYajie/SOG-Net/tree/main/CACE-SOG/model_reduction).

Note that the BSA approximation has been employed in accelerating both classical and quantum many-body LR calculations.^{41,42,47–49} For kernels without an explicit form, as shown in Eq. (10), kernel-independent SOG methods^{45,50} can be employed to construct an appropriate approximation. As shown in our

numerical experiments, this physics-informed initialization may significantly improve convergence speed during the model training.

III. RESULTS

A. Molecular dimers

We first evaluate the CACE-SOG model on a dataset of molecular dimers,⁵¹ which is widely used to benchmark MLIPs in capturing LR interactions across diverse chemical environments.^{7,20} The dataset includes energy and force information computed using the HSE06 hybrid DFT functional with a many-body dispersion correction. The original dataset consists of six types of dimers, each exhibiting distinct ideal power-law decay exponents of the form $1/r^p$. Here, we focus on three representative types: charge-charge (CC, $p = 1$), polar-polar (PP, $p = 3$), and apolar-apolar (AA, $p = 6$). The training set includes ten configurations with intermolecular separations ranging from approximately 5 Å to 12 Å, while the test set consists of three configurations with separations between 12 Å and 15 Å.

We analyze the binding energy curves and atomic forces for the CACE-only, the CACE-LES, and the CACE-SOG models. The CACE representation uses a cutoff $r_{\text{cut}} = 5$ Å, 6 Bessel radial functions, $c = 8$, $\mu_{\text{max}} = 2$, $\nu_{\text{max}} = 2$, $N_{\text{embedding}} = 3$, and one message-passing layer ($T = 1$). Since the two molecules can have a distance beyond the cutoff of $r_{\text{cut}} = 5$ Å, message passing does not help for this task. We keep these settings to match prior CACE-LES work,^{20,34} so the SR part is identical across models and the comparison between CACE-LES and CACE-SOG is fair. For details of these parameters, see Ref. 30. For CACE-LES, we use a one-dimensional ($P = 1$) latent variable and an Ewald smearing factor $\sigma = 1$ Å. The original CACE-LES paper²⁰ uses a four-dimensional latent variable ($P = 4$), while its model test error is comparable with $P = 1$.³⁴ For the CACE-SOG, we use $M = 12$ Gaussians and use the same grid as in CACE-LES. Similar to CACE-LES, we also use a one-dimensional ($P = 1$) latent variable q , which is sufficient for our CACE-SOG model. Figure 2 shows the binding energy prediction errors for each configuration, with the mean absolute errors (MAEs) for energy and forces over all configurations summarized in Table I. MLIPs with LR descriptors are much more accurate than the CACE-only model. The Ewald-based CACE-LES model performs exceptionally well on the CC dimer class, slightly outperforming the CACE-SOG on the test configurations. For dimer types such as PP and AA, whose interactions deviate from the $1/r$ form, the CACE-SOG demonstrates a clear advantage. Its kernel-based representation more effectively captures the LR tails of interactions governed by higher-order power-law decay (e.g., $1/r^3$ and $1/r^6$), highlighting its superior adaptability to diverse physical regimes. The learned deviation of SOG-Net from the BSA initialization reflects both finite- M truncation effects and a data-driven effective correction for deviations of the true LR tail from the ideal decay.

Relative to the DP-SOG,²² the CACE-SOG replaces the fully data-driven DP SR descriptor with the explicit CACE descriptor. This reduces trainable parameters and training-time memory, since backpropagation no longer stores large neighbor-pair activation stacks. Empirically, the CACE-SOG reaches the same or better accuracy with fewer parameters. We report trainable-parameter counts for the tested MLIPs in the first three rows of Table II; the DP-SR and DP-SOG numbers are taken from our previous work.²² On the

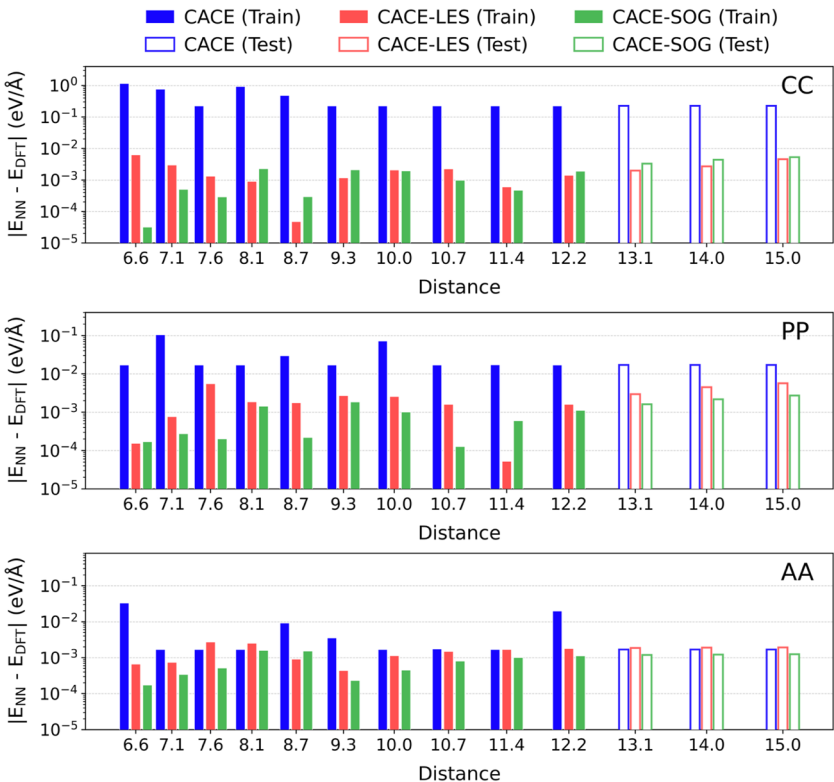


FIG. 2. Binding energies, $E_{NN} - E_{DFT}$ (in eV/Å), are plotted as a function of the distance between the centers of mass of two isolated and relaxed monomers for three dimer classes: charge-charge (CC), polar-polar (PP), and apolar-apolar (AA). Results are shown for the CACE-only, CACE-LES, and CACE-SOG models. Solid and hollow bars indicate the energy differences on the training and testing sets, respectively.

TABLE I. RMSEs of energies (in eV) and forces (in eV/Å) for the CACE-only, CACE-LES, and CACE-SOG models on datasets of CC, PP, and AA dimers.

		CACE	CACE-LES	CACE-SOG
CC	Energy	4.03×10^{-1}	2.72×10^{-3}	2.45×10^{-3}
	Force	8.31×10^{-2}	1.07×10^{-2}	1.16×10^{-2}
PP	Energy	3.16×10^{-2}	2.91×10^{-3}	1.46×10^{-3}
	Force	2.07×10^{-2}	6.15×10^{-3}	4.26×10^{-3}
AA	Energy	1.38×10^{-2}	1.68×10^{-3}	1.04×10^{-3}
	Force	1.95×10^{-3}	9.52×10^{-4}	6.07×10^{-4}

dimer dataset, the CACE-based SR/LR models use about 1/30 and 1/20 as many parameters as the DP-based SR/LR models, respectively. Compared with the CACE-LES, the CACE-SOG uses slightly more parameters since the SOG multiplier introduces an additional $2M$ trainable parameters (M weights and M bandwidths); here, $M = 12$. The small differences in CACE-SOG parameter counts across dimer types come from the number of chemical elements present (scales as $N_{\text{embedding}} \times N_{\text{element}}$), not from the BSA initialization or the target interaction kernel. These results indicate that the SOG-Net layer provides a more flexible LR fit in Fourier space than the fixed Ewald-summation form.

TABLE II. Trainable parameter counts across MLIPs used in this work.

	CACE	CACE-LES	CACE-SOG	DP-SR	DP-SOG
Dimer-CC	13 311	24 016	24 040	431 489	500 242
Dimer-PP	13 311	24 016	24 040	431 489	501 010
Dimer-AA	13 305	24 010	24 034	431 489	498 706
Aqueous KF (MP = 0)	67 234	72 340	72 364	...	...
Aqueous KF (MP = 1)	155 048	212 948	212 972	...	...
$\text{Na}_9\text{Cl}_8^+/\text{Na}_8\text{Cl}_8^+$	7 696	12 796	12 820	...	...
Liquid-vapor	24 572	43 052	43 076	...	...
Water/Pt(111)	67 214	112 514	112 538	...	...

B. Potassium fluoride aqueous solutions

To further assess the accuracy and generalizability of the CACE-SOG, we use a dataset comprising aqueous potassium fluoride (KF) systems³⁴ spanning a wide concentration range from 0 to ~ 2 mol/L. The dataset includes configurations of both bulk electrolytes and interfacial electrolyte-vapor systems. The reference energies and forces of all systems were calculated using the flexible SPC/Fw water model with fixed charges of $-0.8476e$ on oxygen atoms and $+0.4238e$ on hydrogen atoms, together with monovalent potassium and fluoride ions. Figure 3(a) presents the two snapshots in the dataset. The presence of an interface and multiple species introduces nontrivial screening effects, making it a suitable dataset to test our model.

We calculated the results with the CACE-only, CACE-LES, and CACE-SOG, using the same CACE representation for all three MLIPs: 6 Bessel radial functions with $c = 12$, $\mu_{\max} = 3$, $v_{\max} = 3$, $N_{\text{embedding}} = 4$, no message-passing ($T = 0$) or one message-passing layer ($T = 1$), and a cutoff of $r_{\text{cut}} = 4.5$ Å. The setup of the LR component in both the CACE-LES and CACE-SOG follows the same setup as in the molecular dimer tests. Figure 3(b) shows that the CACE-SOG model with a cutoff radius of $r_{\text{cut}} = 4.5$ Å accurately predicts atomic charges as the training set size increases. Figure 3(c) demonstrates that the predicted atomic forces match the reference values closely, achieving a coefficient of determination of $R^2 = 0.999994$. This indicates that the model effectively captures the force landscape across all atomic species. Figures 3(d) and 3(e) compare the performance of the CACE-SOG with the CACE-LES, where the CACE-LES results were obtained from publicly available

data on GitHub.³⁴ Figure 3(d) illustrates that the CACE-SOG significantly improves the MAE over an order of magnitude in charge predictions in most cases with a varying number of samples. This is reasonable because screening in interfacial water makes the LR decay depart from the ideal $1/r$. The CACE-LES uses the classical Ewald summation for a fixed $1/r$ kernel suited to bulk and is less flexible for such tails, while the CACE-SOG learns a data-driven multiplier that adapts to non-Coulombic decay, yielding more accurate inferred charges. The CACE-only model has no latent-variable module, so it cannot predict charges and is omitted in Fig. 3(d). Figure 3(e) displays the MAE of force predictions as a function of the number of training configurations. In Figs. 3(d) and 3(e), adding one message-passing layer changes the results only slightly. This is likely because for $r \geq r_{\text{cut}}$ the LR effects dominate and are already captured by the SOG-Net layer. Regardless of whether message passing is used during the training, the CACE-SOG consistently outperforms the CACE-LES in force prediction accuracy, with MAE reductions of $\sim$ threefold–ninefold as the training set size increases. The gain in force accuracy over the CACE-only model is even more pronounced. These results suggest that the CACE-SOG may offer significant advantages over Ewald-based methods in learning LR interactions for large-scale datasets.

The superior performance of the CACE-SOG can be understood from the nature of interactions in this system. Water molecules are modeled by the SPC/Fw force field, which includes both electrostatic and Lennard-Jones (LJ) potentials for nonbonded interactions. Proper capture of O–O interactions requires an LJ cutoff of 10–12 Å for heterogeneous environments, substantially larger than the typical cutoff for the CACE and other SR descriptors

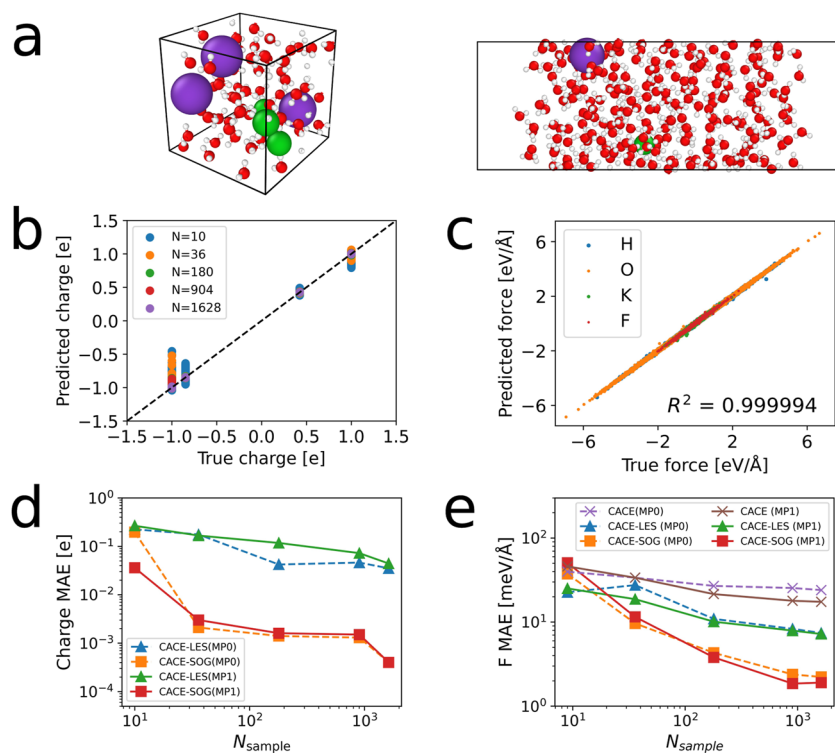


FIG. 3. Results on the aqueous potassium fluoride (KF) dataset. (a) Snapshots of bulk and interfacial systems in the dataset, (b) comparison between true and predicted charges for the CACE-SOG model trained with varying numbers of samples (N), (c) comparison of true and predicted mean absolute errors (MAEs) on forces for each atom type, and [(d) and (e)] MAEs in predicting charges and forces by the CACE-SOG and CACE-LES models without (MP0) and with message passing (MP1) and trained with varying N_{sample} . In panels [(d) and (e)], the additional training samples at each point are randomly drawn from the remaining configurations in the dataset.

(4–7 Å), and increasing the cutoff further increases training cost. Similar issues occur for other pairs such as K–O, K–F, and F–O. Therefore, Ewald summation alone cannot account for LJ contributions beyond the truncation radius. Moreover, for interfacial configurations, water acts as a dielectric medium, and the presence of air–water interfaces introduces depth-dependent screening effects, and LR decay tails significantly deviate from the ideal $1/r$ form. The CACE-SOG accurately simulates this system due to its ability to adaptively capture these LR tails.

C. Dataset with global charge transfer and different charge states

In this section, we test the CACE-SOG model on datasets with global charge transfer or multiple charge states. The first system contains the ionic Na_9Cl_8^+ and Na_8Cl_8^+ clusters when a neutral Na atom is removed,⁶ which exhibits nontrivial charge transfer. The dataset contains 5000 configurations. We used a 90% training and 10% testing split. For CACE representations, we use $r_{\text{cut}} = 5.29$ Å, 6 Bessel radial functions, $c = 8$, $\mu_{\text{max}} = 3$, $\nu_{\text{max}} = 3$, $N_{\text{embedding}} = 2$, and no message passing ($T = 0$). These choices follow prior CACE-LES settings,³⁴ so the SR part is fixed, and we can isolate the effect of the LR layer. Adding one message-passing layer can reduce errors by enlarging the receptive field, but it also increases memory and communication costs. For the CACE-LES, we use a one-dimensional latent variable with a smearing factor of $\sigma = 1.5$ Å. For the CACE-SOG, we use $M = 12$ Gaussians and a grid size of 3 Å, the same as in the CACE-LES.

Table III compares the errors of the CACE-SOG, with and without the initialization strategies described in Sec. II: “w/o” (no special initialization) and “w” (both weights and bandwidths are initialized of the BSA series). We compare against the CACE-only and CACE-LES models. Note that the tested ionic clusters are dominated by electrostatics with a $1/r$ decay, so both the CACE-LES and CACE-SOG can be accurate. The slightly larger error arises when the CACE-SOG is trained from a naïve initialization: the initial LR decay may be mismatched, requiring more epochs, and may converge more slowly. The physics-guided initialization acts like a “normalization” prior on the LR multipliers, improving conditioning and helping prevent overfitting. One can clearly see that the CACE-SOG with the initialization achieves the lowest fitting error among all models.

The learning curves of the pure CACE-only model and the CACE-SOG model with or without the new initialization scheme [i.e., the discretization and proper truncation of Eq. (11)], are shown in Fig. 4. For CACE-SOG, we test three additional strategies besides “w/o”: Type-1 indicates that both weights and bandwidths are initialized from BSA but not trained. Type-2 initializes the bandwidths from BSA, initializes the weights as an all-one vector, and trains

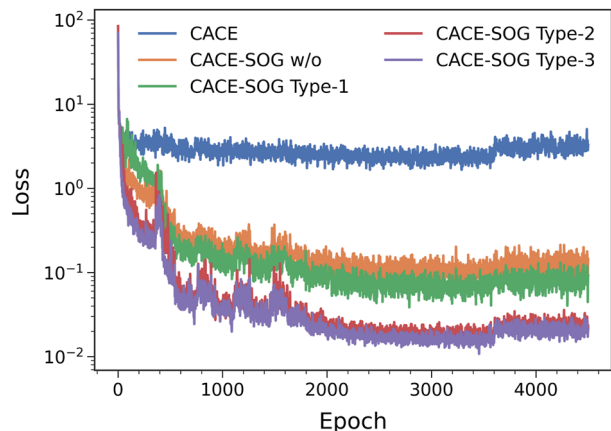


FIG. 4. Learning curves for the ionic Na_9Cl_8^+ and Na_8Cl_8^+ clusters are shown for the CACE-only and CACE-SOG models with different initialization strategies and training protocols for the weights. Details of these strategies and protocols are provided in Sec. III C.

both weights and bandwidths. Type-3 indicates that both weights and bandwidths are initialized according to the strategy by the BSA series. The results demonstrate that the new strategy significantly accelerates error convergence, whereas the model without it converges to lower accuracy. Interestingly, CACE-SOG “w/o” reaches a loss comparable to type-1, which uses the BSA-based initialization but keeps the weights and bandwidths fixed during training. Moreover, type-2 takes roughly 10% more training time than type-3. Finally, type-2 and type-3 achieve nearly identical final accuracy, suggesting that initializing the Gaussian bandwidths is more important than initializing the weights; the weights are comparatively easy to adjust during training. Based on these results, the initialization strategy is used by default in all other numerical tests in this paper.

Next, we benchmark the CACE-SOG on two complex datasets: $\text{Au}_2\text{--MgO}(001)$ and benzotriazole on $\text{Cu}(111)$ (Cu-BTA), containing 5000 and 400 configurations, respectively. The data sources are Ko *et al.*⁶ and Maruf *et al.*⁵² These sets test whether an MLIP can capture charge transfer (e.g., to Au or Cu) and interactions beyond the cutoff. We used a 90% training and 10% testing split. For the CACE representation, we use 6 Bessel radial functions with $c = 12$, $\mu_{\text{max}} = 3$, $\nu_{\text{max}} = 3$, $N_{\text{embedding}} = 4$, and no message-passing layer ($T = 0$). We use a cutoff radius of $r_{\text{cut}} = 5.5$ Å and $r_{\text{cut}} = 5$ Å for $\text{Au}_2\text{--MgO}(001)$ and Cu-BTA, respectively, to match the receptive fields of other benchmark MLIPs. For the CACE-LES, we set $k_c = \pi$ and $\sigma = 1$ Å in the Ewald summation. For CACE-SOG, we use $M = 12$ Gaussians and the same grid size. Table IV reports the energy/force RMSEs and compares the CACE-SOG with charge-constrained atomic cluster expansion (CC-ACE),⁵³ 4G-HDNNP,⁶ NequIP-LR,⁵² the model with polarizable LR potential (Polar-LR),⁵⁴ and the CACE-LES.³⁴ When the LR tail is mainly electrostatic with $1/r$ decay (e.g., Cu-BTA), the CACE-SOG matches the CACE-LES and outperforms other baselines. For systems with more complex LR effects (e.g., $\text{Au}_2\text{--MgO}(001)$), the CACE-SOG also outperforms the CACE-LES. Since the SOG-Net improves on an Ewald-based LR layer, other baselines are likely to benefit when paired with the SOG-Net.

TABLE III. RMSEs of energies (meV/atom) and forces (meV/Å) for the three models on the dataset of the ionic Na_9Cl_8^+ and Na_8Cl_8^+ clusters.

	CACE	CACE-LES	CACE-SOG w/o	CACE-SOG w
Energy	1.58	0.21	0.25	0.12
Force	48.9	9.78	10.46	4.04

TABLE IV. RMSEs of energies (in meV/atom) and forces (in meV/Å) for the SOG-Net in comparison with CC-ACE, 4G-HDNNP, NequIP-LR, Polar-LR, and CACE-LES. Bold marks the best result and underlined marks the second-best.

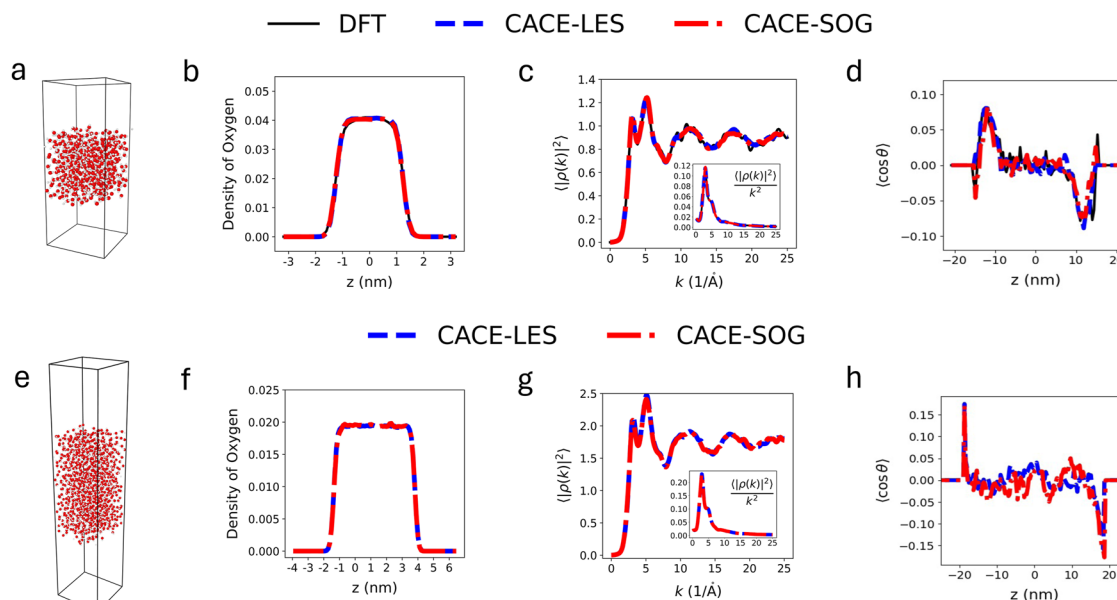
		CC-ACE	4G-HDNNP	NequIP-LR	Polar-LR	CACE-LES	CACE-SOG
Au ₂ -MgO(001)	r_{cut}	6 Å	4.23 Å	5.5 Å	5.5 Å	5.5 Å	5.5 Å
	E	1.63	0.219	0.173	0.14	0.073	<u>0.083</u>
	F	50.27	66.00	21.99	21.0	<u>7.91</u>	5.73
Cu-BTA	r_{cut}	...	...	5 Å	5 Å	5 Å	5 Å
	E	...	...	0.112	0.3	<u>0.074</u>	0.069
	F	...	...	3.48	4.0	3.23	<u>3.24</u>
Ag ₃ ^{+/-} cluster	r_{cut}	6 Å	6 Å	5.29 Å	5.29 Å	5.29 Å	5.29 Å
	E	0.21	1.323	0.821	4.87	0.162	<u>0.171</u>
	F	<u>23.10</u>	31.69	21.21	28.0	29.0	24.5

To test how the CACE-SOG handles different charge states, we validated it on the Ag₃^{+/-}-cluster dataset of Ko *et al.*,⁶ which contains 11 013 configurations of Ag trimers in two charge states. Because the system is small, the whole structure lies within a typical MLIP cutoff, so the main challenge is not LR effects but learning distinct potential energy surfaces for different charge states. Following the last paragraph of Sec. II B, we concatenate the charge state of the system with the SR descriptor and use the combined feature, $D_i \oplus Q$, to predict energy and forces. We use $r_{\text{cut}} = 5.29$ Å, 6 Bessel radial functions with $c = 8$, $\mu_{\text{max}} = 3$, $v_{\text{max}} = 3$, $N_{\text{embedding}} = 1$, and no message-passing layer ($T = 0$). For the CACE-LES, we set $k_c = \pi$ and $\sigma = 1$ Å. For the CACE-SOG, we use $M = 12$ Gaussians and the same

grid size. As shown in the last row of Table IV, the CACE-SOG attains competitive energy and force RMSEs and is slightly more accurate than the CACE-LES, even though LR effects are small in this set. This suggests that the SOG layer learns information beyond the SR part and can also recover some near-field effects lost due to the finite cutoff and descriptor truncation. These results show that the CACE-SOG can handle different charge states.

D. Liquid-vapor interfacial water systems

We now investigate the performance of the CACE-SOG in a liquid-vapor interfacial water dataset.¹⁰ The dataset contains 500 interface configurations with a thickness of 65 Å, each with

**FIG. 5.** Properties of interfacial water predicted using the CACE-LES and CACE-SOG models, in comparison with the reference DFT results. (a) Snapshot of the thinner water slab configuration, (b) the oxygen density distribution, (c) the dipole density correlation function of water molecules, and (d) the average $\cos \theta$ for the angle formed by the water dipole moment and the z -axis. Panels [(e)–(h)] show the corresponding snapshot and results for the thicker water slab.

522 water molecules. The dataset was generated using revPBE0-D3 DFT and has been previously used to train LES and DeePMD models.^{13,20} For the CACE representation, we use a cutoff radius of $r_{\text{cut}} = 5.5 \text{ \AA}$, 6 Bessel radial functions with $c = 12$, $\mu_{\text{max}} = 3$, $\nu_{\text{max}} = 3$, $N_{\text{embedding}} = 3$, and no message-passing layer ($T = 0$). For the LR component, we employ a four-dimensional latent variable representation for both methods. For the CACE-LES, we set the maximum cutoff to $k_c = \pi$ and the smearing factor to $\sigma = 1 \text{ \AA}$ in the Ewald summation. For the CACE-SOG, we use $M = 12$ Gaussians and the same grid size as in the CACE-LES. To test the accuracy of predicting structural properties, we perform 500 ps molecular dynamics simulations in the NVT ensemble at 300 K using the trained CACE-LES and CACE-SOG models, where configurations are dumped with a period of 0.1 ps. The Nosé–Hoover thermostat⁵⁵ is used with a time step of 1 fs. The CACE-LES and CACE-SOG models are trained using random splits of 90% for training and 10% for testing. We then simulate the liquid–vapor interface at 300 K using two different slab thicknesses: a thinner slab with a thickness of 65 Å [Fig. 5(a)], which matches the slab thickness in the training DFT datasets], and a thicker slab with a thickness of 120 Å [Fig. 5(e)]. In Figs. 5(b) and 5(f), we plot the resulting density distributions for the thinner and thicker water slabs, respectively. Both the CACE-LES and CACE-SOG predict similar density profiles. These results show that the CACE-SOG model can transfer to a larger system size without retraining. For the thinner slab, both methods predict a slight boundary repulsion effect, despite the fact that the DFT dataset does not explicitly include such configurations.

In addition, we evaluate the performance of the CACE-SOG in predicting the dielectric properties and orientational order profiles of interfacial water by computing the longitudinal component (i.e., along the z -axis) of the dipole density correlation function (DDCF),

$$D_z(k_z) = \langle |\rho(k_z)|^2 \rangle, \quad (16)$$

where $\rho(k_z)$ is the Fourier transform of the molecular dipole density, and k_z is the z -component of the Fourier modes. We also compute the angle θ between the dipole orientation of water molecules. The dipole correlation function at the long-wavelength limit can be used to determine the dielectric constant of water. As shown in Figs. 5(c) and 5(g), the DDCFs predicted by different MLIPs agree well for most k_z , and $D_z(k_z)/k_z^2$ is nearly constant at long wavelengths due to the symmetry-preserving mean field condition.⁵⁶ In Figs. 5(d) and 5(h), it is shown that the two MLIPs provide slightly different predictions of orientational order, with the CACE-SOG model performing slightly better around the interface. These results demonstrate that the CACE-SOG model accurately captures the dipole layer of water molecules near the liquid–vapor interface, recovering the correct physical polarization behavior.

E. Water/Pt(111) interfacial systems

To further assess the performance difference between the Ewald-based CACE-LES model and the SOG-based CACE-SOG model in more complex scenarios, we study a water/Pt(111) interface system.⁵⁷ Water–metal interfaces are ubiquitous in nature and are relevant to a broad range of technological applications, including corrosion, electrocatalysis, environmental chemistry, and lubrication. The database consists of 48 041 structures, featuring a 3×4 orthogonal Pt(111) slab with a water layer containing 32 water molecules. A snapshot is shown in Fig. 6(a). Reference energies and forces were determined using the Perdew–Burke–Ernzerhof (PBE) functional⁵⁸ combined with the D3 van der Waals correction.⁵⁹ Because both electrostatics and dispersion interactions are included, the effects of the LR component are highly non-negligible. We train the CACE-LES and CACE-SOG models using the same settings as those used for the previous water datasets and then perform molecular dynamics simulations of 1 ns at 350 K within the NVT ensemble, using a Nosé–Hoover thermostat with a time step of 1 fs and a characteristic damping time of 0.1 ps.

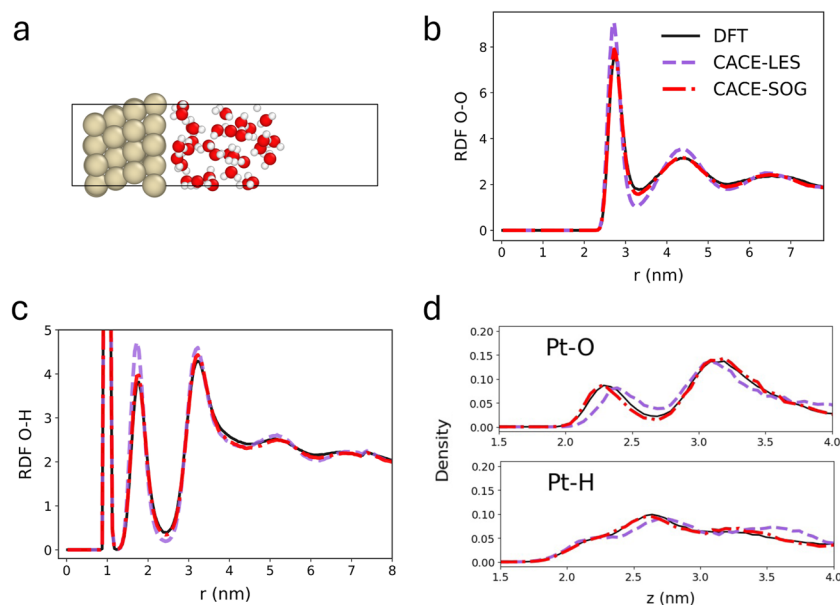


FIG. 6. Results of Pt(111) interfacial water systems by the CACE-LES and CACE-SOG models, in comparison with the reference DFT results. (a) A snapshot of the system, (b) radial distribution function (RDF) of oxygen–oxygen pairs, (c) RDF of oxygen–hydrogen pairs, and (d) planar-averaged density profile of oxygen and hydrogen perpendicular to the interface.

To validate the quality of different MLIPs, we analyzed their ability to reproduce O–O and oxygen–hydrogen (O–H) RDFs. The results of this analysis are shown in Figs. 6(b) and 6(c). Evidently, there is good agreement between the RDFs obtained from our CACE-SOG model and the DFT results. However, although the CACE-LES model provides a reasonable approximation, its peak value deviates noticeably from the DFT reference and CACE-SOG results. This is likely due to the fact that the CACE-LES model is constructed using the classical Ewald summation for the $1/r$ correction, which cannot fully capture the joint LR effects of electrostatic and dispersion interactions.

Next, we study the characterization of the order at the interface. In Fig. 6, we display the planar-averaged density profile of water molecules as a function of distance from the Pt surface. The density of Pt–O exhibits a double-peaked structure at 2–4 Å from the Pt slab, a feature that has also been observed in previous studies of the Pt(111) interfacial water system.^{60,61} One can observe that the CACE-SOG matches the DFT data perfectly and correctly predicts the double-peaked structure and the peak values, demonstrating its high accuracy. This likely stems from its ability to capture LR tails, leading to a better description of interfacial adsorption forces.

IV. CONCLUDING REMARKS

In this paper, we integrate the CACE for the SR descriptor and SOG-Net for the LR descriptor for machine learning of interatomic potentials. This results in a novel, physically interpretable CACE-SOG model that accurately captures interatomic interactions of LR systems. Unlike most previous graph-based MLIP models that heavily rely on the classical Ewald summation to account for LR interactions with a specific $1/r^p$ kernel, the use of SOG-Net enables adaptive learning of LR tails of different decay rates. We also propose an improved initialization for the Fourier multiplier that yields faster convergence and better extrapolation. Systematic experiments, including potassium fluoride aqueous solutions, molecular dimers, ionic clusters, and liquid and interfacial water systems, illustrate that the CACE-SOG model consistently achieves superior accuracy, captures LR charge transfer, predicts partial charges, and recovers structural properties. The results show almost the same accuracy as those in Ji *et al.*,²² where the SOG-Net is used with the SR descriptor of the DP model. Notably, however, the number of trainable parameters in the CACE-SOG is reduced by more than an order of magnitude. Even when compared to the Ewald-based CACE-LES model, the CACE-SOG achieves higher accuracy while employing nearly the same number of parameters (see Table II). These results demonstrate the strong performance of the CACE-SOG and show that SOG-Net is a general module for LR interactions with variable decay tails.

In real systems, the potential energy surfaces of different systems, where LR interactions play a central role, may exhibit distinct far-field decay behaviors for each atom type. In this context, developing reliable foundation models that capture LR physics is crucial. Several challenges remain in this direction. Our future work includes extending the current model to higher-order interactions beyond two-body terms and testing it on larger datasets generated by advanced methods with quantum-mechanical accuracy. Ultimately, our goal is to learn from the data to identify the types of LR

interaction and their decay properties, providing users with a clearer physical picture.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) accompanying this article contains detailed derivations of Eqs. (10)–(13) and a description of the model reduction technique.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant Nos. 12325113, 12401570, 12426304, 124B2023, and 12350710181) and the Science and Technology Commission of Shanghai Municipality (Grant No. 23JC1402300). The work of J.L. is partially supported by the China Postdoctoral Science Foundation (Grant No. 2024M751948). The authors would like to thank the support from the SJTU Kunpeng & Ascend Center of Excellence.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yajie Ji: Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jiuyang Liang:** Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Supervision (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Zhenli Xu:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The CACE-SOG package, training sets, and training scripts are publicly available at <https://github.com/DuktigYajie/SOG-Net>.

REFERENCES

- ¹O. T. Unke, S. Chmiela, H. E. Sauceda, M. Gastegger, I. Poltavsky, K. T. Schütt, A. Tkatchenko, and K.-R. Müller, *Chem. Rev.* **121**, 10142 (2021).
- ²P. Friederich, F. Häse, J. Proppe, and A. Aspuru-Guzik, *Nat. Mater.* **20**, 750 (2021).
- ³E. Prodan and W. Kohn, *Proc. Natl. Acad. Sci. U. S. A.* **102**, 11635 (2005).
- ⁴N. Fedik, R. Zubatyuk, M. Kulichenko, N. Lubbers, J. S. Smith, B. Nebgen, R. Messerly, Y. W. Li, A. I. Boldyrev, K. Barros, O. Isayev, and S. Tretiak, *Nat. Rev. Chem.* **6**, 653 (2022).
- ⁵C. Zhang, M. F. Calegari Andrade, Z. K. Goldsmith, A. S. Raman, Y. Li, P. M. Piaggi, X. Wu, R. Car, and A. Selloni, *Nat. Commun.* **15**, 10270 (2024).

- ⁶T. W. Ko, J. A. Finkler, S. Goedecker, and J. Behler, *Nat. Commun.* **12**, 398 (2021).
- ⁷K. K. Huguenin-Dumittan, P. Loche, N. Haoran, and M. Ceriotti, *J. Phys. Chem. Lett.* **14**, 9612 (2023).
- ⁸H.-X. Zhou and X. Pang, *Chem. Rev.* **118**, 1691 (2018).
- ⁹J. Yuan and H. Tanaka, *Nat. Commun.* **16**, 1517 (2025).
- ¹⁰S. P. Niblett, M. Galib, and D. T. Limmer, *J. Chem. Phys.* **155**, 164101 (2021).
- ¹¹O. T. Unke, S. Chmiela, M. Gastegger, K. T. Schütt, H. E. Sauceda, and K.-R. Müller, *Nat. Commun.* **12**, 7273 (2021).
- ¹²A. K. Rappe and W. A. Goddard III, *J. Phys. Chem.* **95**, 3358 (1991).
- ¹³L. Zhang, H. Wang, M. C. Muniz, A. Z. Panagiotopoulos, R. Car, and W. E, *J. Chem. Phys.* **156**, 124107 (2022).
- ¹⁴A. Gao and R. C. Remsing, *Nat. Commun.* **13**, 1572 (2022).
- ¹⁵T. Darden, D. York, and L. Pedersen, *J. Chem. Phys.* **98**, 10089 (1993).
- ¹⁶M. Gubler, J. A. Finkler, M. R. Schafer, J. Behler, and S. Goedecker, *J. Chem. Theory Comput.* **20**, 7264 (2024).
- ¹⁷A. Grisafi and M. Ceriotti, *J. Chem. Phys.* **151**, 204105 (2019).
- ¹⁸A. Kosmala, J. Gasteiger, N. Gao, and S. Günnemann, in *International Conference on Machine Learning* (PMLR, 2023), pp. 17544–17563.
- ¹⁹W. Kohn, *Phys. Rev. Lett.* **76**, 3168 (1996).
- ²⁰B. Cheng, *npj Comput. Mater.* **11**, 80 (2025); GitHub repository <https://github.com/BingqingCheng/cace-lr-fit>.
- ²¹T. W. Ko and S. P. Ong, *Nat. Comput. Sci.* **3**, 998 (2023).
- ²²Y. Ji, J. Liang, and Z. Xu, *Phys. Rev. Lett.* **135**, 178001 (2025).
- ²³L. Zhang, J. Han, H. Wang, R. Car, and W. E, *Phys. Rev. Lett.* **120**, 143001 (2018).
- ²⁴A. Dutt and V. Rokhlin, *SIAM J. Sci. Comput.* **14**, 1368 (1993).
- ²⁵L. Greengard and J.-Y. Lee, *SIAM Rev.* **46**, 443 (2004).
- ²⁶I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csányi, *Advances in Neural Information Processing Systems* (Curran Associates, Inc., 2022), Vol. 35, p. 11423.
- ²⁷A. V. Shapeev, *Multiscale Model. Simul.* **14**, 1153 (2016).
- ²⁸S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, and B. Kozinsky, *Nat. Commun.* **13**, 2453 (2022).
- ²⁹R. Drautz, *Phys. Rev. B* **99**, 014104 (2019).
- ³⁰B. Cheng, *npj Comput. Mater.* **10**, 157 (2024).
- ³¹R. Jacobs, D. Morgan, S. Attarian, J. Meng, C. Shen, Z. Wu, C. Y. Xie, J. H. Yang, N. Artrith, B. Blaiszik *et al.*, *Curr. Opin. Solid State Mater. Sci.* **35**, 101214 (2025).
- ³²B. Kozinsky, A. Musaelian, A. Johansson, and S. Batzner, in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis* (Association for Computing Machinery, 2023), pp. 1–12.
- ³³P. P. Ewald, *Ann. Phys.* **369**, 253 (1921).
- ³⁴D. S. King, D. Kim, P. Zhong, and B. Cheng, *Nat. Commun.* **16**, 8763 (2025).
- ³⁵P. Zhong, D. Kim, D. S. King, and B. Cheng, *arXiv:2504.05169* (2025).
- ³⁶J. Behler and M. Parrinello, *Phys. Rev. Lett.* **98**, 146401 (2007).
- ³⁷A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi, *Phys. Rev. Lett.* **104**, 136403 (2010).
- ³⁸B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, and G. Ceder, *Nat. Mach. Intell.* **5**, 1031 (2023).
- ³⁹A. H. Barnett, J. Magland, and L. af Klinteberg, *SIAM J. Sci. Comput.* **41**, C479 (2019).
- ⁴⁰G. Beylkin and L. Monzón, *Appl. Comput. Harmonic Anal.* **28**, 131 (2010).
- ⁴¹C. Predescu, A. K. Lerer, R. A. Lippert, B. Towles, J. P. Grossman, R. M. Dirks, and D. E. Shaw, *J. Chem. Phys.* **152**, 084113 (2020); *arXiv:1911.01377*.
- ⁴²J. Liang, Z. Xu, and Q. Zhou, *Appl. Comput. Harmonic Anal.* **77**, 101759 (2025).
- ⁴³P. Benner, S. Gugercin, and K. Willcox, *SIAM Rev.* **57**, 483 (2015).
- ⁴⁴K. Xu and S. Jiang, *J. Sci. Comput.* **55**, 16 (2013).
- ⁴⁵Z. Gao, J. Liang, and Z. Xu, *J. Sci. Comput.* **93**, 40 (2022).
- ⁴⁶Y. Lin, Z. Xu, Y. Zhang, and Q. Zhou, *Phys. Rev. E* **112**, 015302 (2025).
- ⁴⁷J. Kaye, Z. Huang, H. U. R. Strand, and D. Golež, *Phys. Rev. X* **14**, 031034 (2024).
- ⁴⁸J. Liang, Z. Xu, and Q. Zhou, *SIAM J. Sci. Comput.* **45**, B591 (2023).
- ⁴⁹Q. Zhou, T. Wu, J. Liu, Q. Sun, H. Xie, and Z. Xu, *arXiv:2508.10454* (2025).
- ⁵⁰L. Greengard, S. Jiang, and Y. Zhang, *SIAM J. Sci. Comput.* **40**, A3733 (2018).
- ⁵¹L. A. Burns, J. C. Faver, Z. Zheng, M. S. Marshall, D. G. A. Smith, K. Vanommes-laeghe, A. D. MacKerell, K. M. Merz, and C. D. Sherrill, *J. Chem. Phys.* **147**, 161727 (2017).
- ⁵²M. U. Maruf, S. Kim, and Z. Ahmad, *J. Phys. Chem. Lett.* **16**, 9078 (2025).
- ⁵³M. Rinaldi, A. Bochkarev, Y. Lysogorskiy, and R. Drautz, *Phys. Rev. Mater.* **9**, 033802 (2025).
- ⁵⁴R. Gao, C. Yam, J. Mao, S. Chen, G. Chen, and Z. Hu, *arXiv:2410.13820* (2025).
- ⁵⁵S. Nosé, *J. Chem. Phys.* **81**, 511 (1984).
- ⁵⁶Z. Hu, *J. Chem. Phys.* **156**, 034111 (2022).
- ⁵⁷A. E. Mikkelsen, J. Schiøtz, T. Vegge, and K. W. Jacobsen, *J. Chem. Phys.* **155**, 224701 (2021).
- ⁵⁸J. P. Perdew, K. Burke, and M. Ernzerhof, *Phys. Rev. Lett.* **77**, 3865 (1996).
- ⁵⁹S. Grimme, J. Antony, S. Ehrlich, and H. Krieg, *J. Chem. Phys.* **132**, 154104 (2010).
- ⁶⁰H. H. Kristoffersen, T. Vegge, and H. A. Hansen, *Chem. Sci.* **9**, 6912 (2018).
- ⁶¹D. T. Limmer, A. P. Willard, P. Madden, and D. Chandler, *Proc. Natl. Acad. Sci. U. S. A.* **110**, 4200 (2013).